Subject Code	**AF5364**
Subject Title	**Quantitative Methods for Accounting and Finance**
Credit Value	3
Level	5
Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	Recommended Background Knowledge: Undergraduate level statistical analysis; quantitative analysis; and microeconomics.
Objectives	This subject covers the basic concepts and techniques of classical econometrics, such as sampling theory, probability theory, hypothesis testing, regressions, etc. Considerable attention is devoted to the applications of the concepts and techniques to accounting and finance. We will review basic financial mathematics and some advanced statistical techniques will also be briefly introduced. This subject is also designed for those who wish to take the Chartered Financial Analysts (CFA) examinations. This subject contributes to the achievement of the MSc in Accounting and Finance Analytics programme learning outcomes by enabling students to understand the fundamental quantitative and technological methods in accounting and finance (MSc AFA Programme Outcome 2).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Develop a systematic understanding of the fundamental statistical and econometric concepts and methodologies; b. Apply the concepts and methodologies to explain issues related to accounting and finance; and c. Develop ability to resolve real world accounting and finance problems by applying the quantitative methods to data analysis.
Subject Synopsis/ Indicative Syllabus	**Basic Financial Mathematics (Review)** Compounding and discounting; present value and future value calculations; annuities and perpetuities; dollar and time-weighted rate of return.

	Basic Statistics Concepts Types of statistical data; measures of central tendency and dispersion. **Probability Concepts** Basic concepts of probability; random variables and probability; probability theorems; covariance and correlation; expected value and variance; probability distributions. **Sampling and Estimation** Random sampling and sampling distributions; point and interval estimates; confidence intervals. **Hypothesis Testing and Statistical Inference** The concepts of hypothesis testing; types of hypothesis testing; analysis of variance. **Regression Analysis** Linear regression and correlation; multiple regression analysis.
Teaching/Learning Methodology	Concepts and techniques will be introduced through lectures. Students are required to apply the knowledge and skills in doing exercises and project. The use of relevant computer package is required.

Assessment Methods in Alignment with Intended Learning Outcomes	(see table below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c			
1. Class Participation	10%	✓	✓	✓			
2. Homework	20%	✓	✓	✓			
3. Project	20%	✓	✓	✓			
4. Final Examination	50%	✓	✓	✓			
Total	100 %						

Note: The specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers.

Student Study Effort Expected		
	Class contact:	
	Lectures / Seminars	39 Hrs.
	Other student study effort:	
	Reading materials/textbook and working on exercises, depending on each student's background.	78 Hrs.
	Total student study effort	117 Hrs.

Reading List and References	
	Quantitative Investment Analysis, by Richard Armand Defusco, Dennis W. McLeavey, Jerald E. Pinto, David E. Runkle, 4th edition, John Wiley & Sons, Inc. Econometric Methods, 4th edition by Jack Johnston and John DiNardo Some additional readings will be distributed in class.